

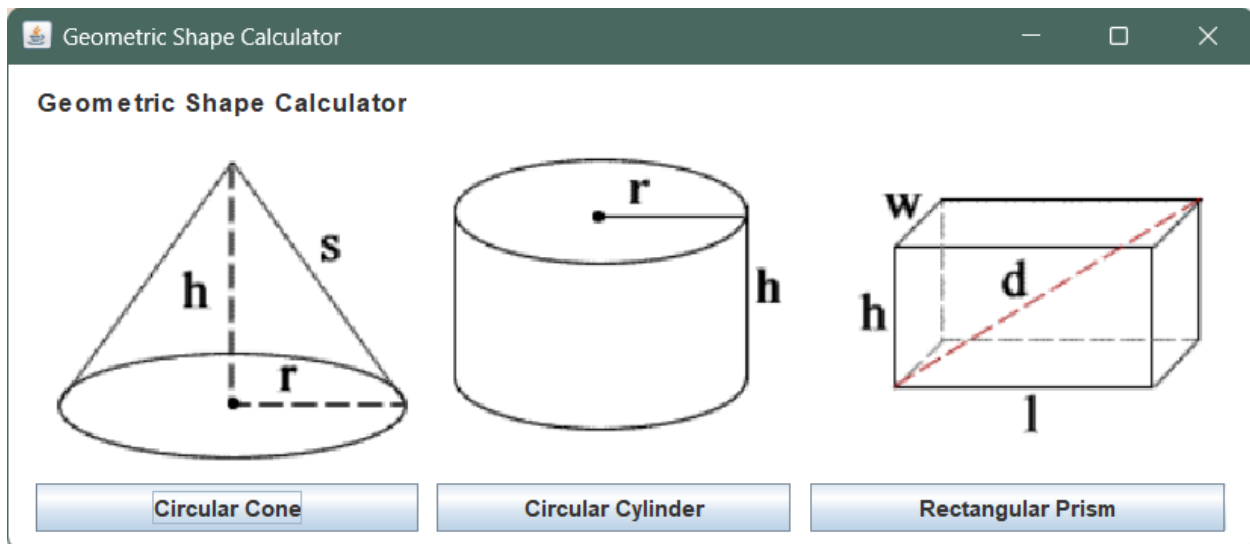
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Lab #10

ฝึกการใช้ GUI และ JAR

Question #1

1. จาก Java Project ชื่อว่า **GeometricShapeCalculator** ในเนื้อหาช่วงบรรยาย
ให้นักศึกษาพัฒนาส่วนที่เหลือให้ครบสมบูรณ์
2. สร้าง GUI Form ในรูปทรงส่วนที่เหลือให้ครบถ้วน ได้แก่ Circular Cylinder และ Rectangular Prism
โดยในส่วนของ Sub Form ทั้ง 2 นั้นให้ออกแบบได้ตามสะดวก



คำตอบ #1

Source Code

```
GeometricShapes.java × CircularCylinder.java × RectangularPrism.java × CircularCone.java

1 package it.util.shapes;
2
3 public interface GeometricShapes { 3 usages 3 implementations
4     public double getVolume(); no usages 3 implementations
5     public double getTotalSurfaceArea(); no usages 3 implementations
6 }
7
```

```
GeometricShapes.java × CircularCylinder.java × RectangularPrism.java × CircularCone.java

1 package it.util.shapes;
2
3 public class CircularCylinder implements GeometricShapes { no usages
4     private double radius; 8 usages
5     private double height; 4 usages
6
7     public CircularCylinder(double r, double h) { no usages
8         this.radius = r;
9         this.height = h;
10    }
11
12    @Override no usages
13    public double getVolume() {
14        double v = Math.PI * radius * radius * height;
15        return v;
16    }
17
18    @Override no usages
19    public double getTotalSurfaceArea() {
20        double ts = getLateralSurfaceArea() + getTopSurfaceArea() + getBottomSurfaceArea();
21        return ts;
22    }
23
24    public double getLateralSurfaceArea() { 1 usage
25        double l = 2 * Math.PI * radius * height;
26        return l;
27    }
28}
```

```

28
29     public double getTopSurfaceArea() { 1 usage
30         double t = Math.PI * radius * radius;
31         return t;
32     }
33
34     public double getBottomSurfaceArea() { 1 usage
35         double b = Math.PI * Math.pow(radius,2);
36         return b;
37     }
38
39     public double getRadius() { no usages
40         return this.radius;
41     }
42
43     public double getHeight() { no usages
44         return this.height;
45     }

```

```

GeometricShapes.java × CircularCylinder.java RectangularPrism.java × CircularCone.java
1 package it.util.shapes;
2
3 public class RectangularPrism implements GeometricShapes { no usages
4     private double width; 6 usages
5     private double height; 6 usages
6     private double length; 6 usages
7
8     public RectangularPrism(double w, double h, double l) { no usages
9         this.width = w;
10        this.height = h;
11        this.length = l;
12    }
13
14    @Override no usages
15    public double getVolume() {
16        return length*width*height;
17    }
18
19    @Override no usages
20    public double getTotalSurfaceArea() {
21        double ts = 2 * ((length*width)+(length*height)+(width*height));
22        return ts;
23    }
24
25    public double getDiagonal() { no usages
26        double d = Math.sqrt(Math.pow(length,2)+Math.pow(width,2)+Math.pow(height,2));
27        return d;
28    }

```

```

29
30     public double getWidth() { no usages
31         |     return width;
32     }
33
34     public double getHeight() { no usages
35         |     return height;
36     }
37
38     public double getLength() { no usages
39         |     return length;
40     }
41

```

GeometricShapes.java × CircularCylinder.java RectangularPrism.java CircularCone.java ×

```

1     package it.util.shapes;
2
3     public class CircularCone implements GeometricShapes { no usages
4         |     private double radius; 6 usages
5         |     private double height; 4 usages
6
7         public CircularCone (double r, double h) { no usages
8             |     this.radius = r;
9             |     this.height = h;
10        }
11
12        @Override no usages
13        public double getVolume() {
14            |     return (1.0/3) * (Math.PI * Math.pow(radius,2) * height);
15        }
16
17        @Override no usages
18        public double getTotalSurfaceArea() {
19            |     return getLateralSurfaceArea()+getBaseSurfaceArea();
20        }
21
22        public double getLateralSurfaceArea() { 1 usage
23            |     return Math.PI * radius * getSlantHeight();
24        }
25
26        public double getBaseSurfaceArea() { 1 usage
27            |     return Math.PI * Math.pow(radius,2);
28        }

```

```

29
30     public double getSlantHeight() { 1 usage
31         return Math.sqrt(Math.pow(radius,2)+Math.pow(height,2));
32     }
33
34     public double getRadius() { no usages
35         return radius;
36     }
37
38     public double getHeight() { no usages
39         return height;
40     }
41

```

MainForm.java ×

```

1  import javax.swing.*;
2  import java.awt.*;
3  import java.awt.event.ActionEvent;
4  import java.awt.event.ActionListener;
5  import java.awt.event.MouseAdapter;
6  import java.awt.event.MouseEvent;
7
8  // JFrame = หน้าจอ windows
9  public class MainForm extends JFrame {
10     private JPanel mainPanel; 2 usages
11     private JButton circularConeButton; 2 usages
12     private JButton circularCylinderButton; 2 usages
13     private JButton rectangularPrismButton; 2 usages
14     private JLabel circularConeLabel; 4 usages
15     private JLabel circularCylinderLabel; 4 usages
16     private JLabel rectangularPrismLabel; 4 usages
17
18     public static final String CIRCULARCONE = "CIRCULARCONE"; 3 usages
19     public static final String CIRCULARCYLINDER = "CIRCULARCYLINDER"; 3 usages
20     public static final String RECTANGULARPRISM = "RECTANGULARPRISM"; 3 usages
21

```

```

22     public MainForm() { 1 usage
23         initialFormProperty();
24         circularConeLabel.addMouseListener(new MouseAdapter() {
25             @Override
26             public void mouseClicked(MouseEvent e) {
27                 openForm(MainForm.CIRCULARCONE);
28             }
29         });
30         circularCylinderLabel.addMouseListener(new MouseAdapter() {
31             @Override
32             public void mouseClicked(MouseEvent e) {
33                 openForm(MainForm.CIRCULARCYLINDER);
34             }
35         });
36         rectangularPrismLabel.addMouseListener(new MouseAdapter() {
37             @Override
38             public void mouseClicked(MouseEvent e) {
39                 openForm(MainForm.RECTANGULARPRISM);
40             }
41         });
42         circularConeButton.addActionListener(new ActionListener() {
43             @Override
44             public void actionPerformed(ActionEvent e) {
45                 openForm(MainForm.CIRCULARCONE);
46             }
47         });

```

```

48         circularCylinderButton.addActionListener(new ActionListener() {
49             @Override
50             public void actionPerformed(ActionEvent e) {
51                 openForm(MainForm.CIRCULARCYLINDER);
52             }
53         });
54         rectangularPrismButton.addActionListener(new ActionListener() {
55             @Override
56             public void actionPerformed(ActionEvent e) {
57                 openForm(MainForm.RECTANGULARPRISM);
58             }
59         });
60     }
61
62     @ private void openForm(String formName) { 6 usages
63         switch (formName) {
64             case MainForm.CIRCULARCONE:
65                 CircularConeForm circularConeForm = new CircularConeForm();
66                 circularConeForm.show();
67                 break;
68             case MainForm.CIRCULARCYLINDER:
69                 CircularCylinderForm circularCylinderForm = new CircularCylinderForm();
70                 circularCylinderForm.show();
71                 break;
72             case MainForm.RECTANGULARPRISM:
73                 RectangularPrismForm rectangularPrismForm = new RectangularPrismForm();
74                 rectangularPrismForm.show();
75                 break;

```

```

76     }
77 }
78
79 private void displayError(String msg) { 1 usage
80     JOptionPane.showMessageDialog(parentComponent null, msg, title: "Error", JOptionPane.ERROR_MESSAGE);
81 }
82
83 private void initialFormProperty() { 1 usage
84     try {
85         circularConeLabel.setIcon(new ImageIcon(this.getClass().getResource(name: "/resources/circularcone.png")));
86         circularCylinderLabel.setIcon(new ImageIcon(this.getClass().getResource(name: "/resources/circularcylinder.png")));
87         rectangularPrismLabel.setIcon(new ImageIcon(this.getClass().getResource(name: "/resources/rectangularprism.png")));
88         // set cursor hover on images คลิกที่รูปดาว
89         circularConeLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
90         circularCylinderLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
91         rectangularPrismLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
92     } catch (Exception e) {
93         displayError(e.getMessage());
94         System.exit(status: 1);
95     }
96 }
97

```

```

98 public static void main(String[] args) {
99     MainForm mainForm = new MainForm();
100     mainForm.setTitle("Geometric Shape Calculator");
101     mainForm.setContentPane(mainForm.mainPanel);
102     mainForm.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // ออกจาก windows
103     mainForm.pack();
104     mainForm.setVisible(true); // ให้ออกมา
105 }
106 }

```

```

CircularConeForm.java x CircularCylinderForm.java RectangularPrismForm.java x
1 import it.util.shapes.CircularCylinder;
2 import it.util.shapes.RectangularPrism;
3
4 import javax.swing.*;
5 import java.awt.*;
6 import java.awt.event.ActionEvent;
7 import java.awt.event.ActionListener;
8
9 public class RectangularPrismForm { 4 usages
10     private JPanel rectangularPrismPanel; 2 usages
11     private JLabel rectangularPrismLabel; 2 usages
12     private JTextField textWidth; 2 usages
13     private JTextField textHeight; 2 usages
14     private JTextField textLength; 2 usages
15     private JButton calculateButton; 2 usages
16     private JButton closeButton; 2 usages
17     private JDialog dialog; 7 usages
18
19     public RectangularPrismForm() { 1 usage
20         rectangularPrismLabel.setIcon(new ImageIcon(this.getClass().getResource(name: "/resources/rectangularprism2.png")));
21         dialog = new JDialog((Frame) null, title: "Rectangular Prism Calculator", modal: true);
22         dialog.setContentPane(rectangularPrismPanel);
23         dialog.pack();
24         dialog.setDefaultCloseOperation(WindowConstants.DISPOSE_ON_CLOSE);
25         calculateButton.addActionListener(new ActionListener() {
26             @Override
27             public void actionPerformed(ActionEvent e) {
28                 doCalculate();
29             }
30 });

```

```

31         closeButton.addActionListener(new ActionListener() {
32             @Override
33             public void actionPerformed(ActionEvent e) {
34                 dispose();
35             }
36         });
37     }
38
39     private void doCalculate() { 1 usage
40         try {
41             RectangularPrism rc = new RectangularPrism(Double.parseDouble(textWidth.getText()),
42                                                         Double.parseDouble(textHeight.getText()),
43                                                         Double.parseDouble(textLength.getText()));
44             double volume = rc.getVolume();
45             double diagonal = rc.getDiagonal();
46             double totalSurfaceArea = rc.getTotalSurfaceArea();
47             String result = "Volume = " + volume + "\n" +
48                             "Diagonal = " + diagonal + "\n" +
49                             "Total Surface Area = " + totalSurfaceArea;
50             displayResult(result, "Result of Circular Cylinder Shape", JOptionPane.INFORMATION_MESSAGE);
51         } catch (NumberFormatException ex) {
52             displayError("Error", "Please enter number Only!");
53         }
54     }
55
56     public void displayResult(String resultMsg, String title, int type) { 1 usage
57         JOptionPane.showMessageDialog(parentComponent, null, resultMsg, title, type);
58     }

```

```

59
60     public void displayError(String title, String errorMessage) { 1 usage
61         JOptionPane.showMessageDialog(parentComponent, null, errorMessage, title, JOptionPane.ERROR_MESSAGE);
62     }
63
64     public void show() { 1 usage
65         dialog.setVisible(true);
66     }
67
68     public void dispose() { 1 usage
69         dialog.setVisible(false);
70         dialog.dispose();
71     }
72 }
73

```



```
CircularConeForm.java x CircularCylinderForm.java x
1 import it.util.shapes.CircularCylinder;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7
8 public class CircularCylinderForm { 4 usages
9     private JTextField textRadius1; 2 usages
10    private JTextField textHeight1; 2 usages
11    private JButton calculateButton; 2 usages
12    private JButton closeButton; 2 usages
13    private JLabel circularCylinderLabel; 2 usages
14    private JPanel circularCylinderPanel; 2 usages
15    private JDialog dialog; 7 usages
16
17
18    public CircularCylinderForm() { 1 usage
19        circularCylinderLabel.setIcon(new ImageIcon(this.getClass().getResource("/resources/circularcylinder2.png")));
20        dialog = new JDialog((Frame) null, "Circular Cone Calculator", modal: true);
21        dialog.setContentPane(circularCylinderPanel);
22        dialog.pack();
23        dialog.setDefaultCloseOperation(WindowConstants.DISPOSE_ON_CLOSE);
24        calculateButton.addActionListener(new ActionListener() {
25            @Override
26            public void actionPerformed(ActionEvent e) {
27                doCalculate();
28            }
29        });
30    }
```

```
30        closeButton.addActionListener(new ActionListener() {
31            @Override
32            public void actionPerformed(ActionEvent e) {
33                dispose();
34            }
35        });
36    }
37
38    private void doCalculate() { 1 usage
39        try {
40            CircularCylinder circularCylinder = new CircularCylinder(Double.parseDouble(textRadius1.getText()),
41                                                                    Double.parseDouble(textHeight1.getText()));
42            double volume = circularCylinder.getVolume();
43            double lateralSurfaceArea = circularCylinder.getLateralSurfaceArea();
44            double topSurfaceArea = circularCylinder.getTopSurfaceArea();
45            double bottomSurfaceArea = circularCylinder.getBottomSurfaceArea();
46            double totalSurfaceArea = circularCylinder.getTotalSurfaceArea();
47            String result = "Volume = " + volume + "\n" +
48                            "Lateral Surface Area = " + lateralSurfaceArea + "\n" +
49                            "Top Surface Area = " + topSurfaceArea + "\n" +
50                            "Bottom Surface Area = " + bottomSurfaceArea + "\n" +
51                            "Total Surface Area = " + totalSurfaceArea;
52            displayResult(result, "Result of Circular Cylinder Shape", JOptionPane.INFORMATION_MESSAGE);
53        } catch (NumberFormatException ex) {
54            displayError("Error", "Please enter number Only!");
55        }
56    }
```

```

57
58     public void displayResult(String resultMsg, String title, int type) { 1 usage
59         JOptionPane.showMessageDialog(parentComponent: null, resultMsg, title, type);
60     }
61
62     public void displayError(String title, String errorMessage) { 1 usage
63         JOptionPane.showMessageDialog(parentComponent: null, errorMessage, title, JOptionPane.ERROR_MESSAGE);
64     }
65
66     public void show() { 1 usage
67         dialog.setVisible(true);
68     }
69
70     public void dispose() { 1 usage
71         dialog.setVisible(false);
72         dialog.dispose();
73     }
74 }

```

CircularConeForm.java ×

```

1  import it.util.shapes.CircularCone;
2
3  import javax.swing.*;
4  import java.awt.*;
5  import java.awt.event.ActionEvent;
6  import java.awt.event.ActionListener;
7
8  public class CircularConeForm { 4 usages
9      private JTextField textHeight; 2 usages
10     private JTextField textRadius; 2 usages
11     private JButton calculateButton; 2 usages
12     private JButton closeButton; 2 usages
13     private JLabel circularConeLabel; 2 usages
14     private JPanel circularConePanel; 2 usages
15     private JDialog frame; 7 usages
16
17     public CircularConeForm() { 1 usage
18         circularConeLabel.setIcon(new ImageIcon(this.getClass().getResource("/resources/circularcone2.png")));
19         frame = new JDialog((Frame) null, title: "Circular Cone Calculator", modal: true);
20         frame.setContentPane(circularConePanel);
21         frame.pack();
22         frame.setDefaultCloseOperation(WindowConstants.DISPOSE_ON_CLOSE);
23         closeButton.addActionListener(new ActionListener() {
24             @Override
25             public void actionPerformed(ActionEvent e) {
26                 dispose();
27             }
28         });

```

```

29         calculateButton.addActionListener(new ActionListener() {
30             @Override
31             public void actionPerformed(ActionEvent e) {
32                 doCalculate();
33             }
34         });
35     }
36
37     private void doCalculate() { 1usage
38         try {
39             CircularCone cc = new CircularCone(Double.parseDouble(textRadius.getText()),
40                 Double.parseDouble(textHeight.getText()));
41             double volume = cc.getVolume(); // call calculate volume method for relative shape in JAR
42             double slantHeight = cc.getSlantHeight(); // call calculate slant method for relative shape in JAR
43             double lateralSurfaceArea = cc.getLateralSurfaceArea(); // call calculate lateral surface area method for relative shape in JAR
44             double baseSurfaceArea = cc.getBaseSurfaceArea(); // call calculate base surface area method for relative shape in JAR
45             double totalSurfaceArea = cc.getTotalSurfaceArea(); // call calculate total surface area method for relative shape in JAR
46             String result = "Volume = " + volume + "\n" +
47                 "Slant Height = " + slantHeight + "\n" +
48                 "Lateral Surface Area = " + lateralSurfaceArea + "\n" +
49                 "Base Surface Area = " + baseSurfaceArea + "\n" +
50                 "Total Surface Area = " + totalSurfaceArea;
51             displayResult(result, title: "Result of Circular Cone Shape", JOptionPane.INFORMATION_MESSAGE);
52         } catch (NumberFormatException ex) {
53             displayError(title: "Error", errorMessage: "Please enter number Only!");
54         }
55     }

```

```

56
57     public void displayResult(String resultMsg, String title, int type) { 1usage
58         JOptionPane.showMessageDialog(parentComponent: null, resultMsg, title, type);
59     }
60
61     public void displayError(String title, String errorMessage) { 1usage
62         JOptionPane.showMessageDialog(parentComponent: null, errorMessage, title, JOptionPane.ERROR_MESSAGE);
63     }
64
65
66     public void show() { 1usage
67         frame.setVisible(true);
68     }
69
70     public void dispose() { 1usage
71         frame.setVisible(false);
72         frame.dispose();
73     }
74 }

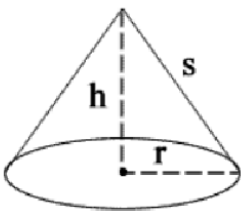
```

ผลการทำงาน

Circular Cone

Circular Cone Calculator

Circular Cone



- Volume = $(1/3)\pi r^2 h$
- Slant Height = $\sqrt{r^2 + h^2}$
- Lateral Surface Area = $\pi r s = \pi r \sqrt{r^2 + h^2}$
- Base Surface Area = πr^2
- Total Surface Area
= L + B = $\pi r s + \pi r^2 = \pi r(s + r) = \pi r(r + \sqrt{r^2 + h^2})$

Radius: 4
Height: 2

Calculate

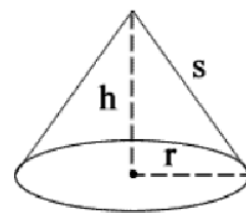
Result of Circular Cone Shape

Volume = 33.510321638291124
Slant Height = 4.47213595499958
Lateral Surface Area = 56.198517848325814
Base Surface Area = 50.26548245743669
Total Surface Area = 106.46400030576251

OK

Circular Cone Calculator

Circular Cone



- Volume = $(1/3)\pi r^2 h$
- Slant Height = $\sqrt{r^2 + h^2}$
- Lateral Surface Area = $\pi r s = \pi r \sqrt{r^2 + h^2}$
- Base Surface Area = πr^2
- Total Surface Area
= L + B = $\pi r s + \pi r^2 = \pi r(s + r) = \pi r(r + \sqrt{r^2 + h^2})$

Radius: yy
Height: gg

Calculate

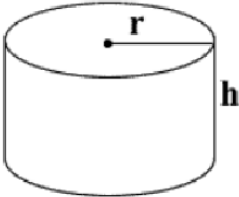
Error

Please enter number Only!

OK

Circular Cylinder

Circular Cylinder



- Volume = $\pi r^2 h$
- Lateral Surface Area = $2\pi r h$
- Top Surface Area = πr^2
- Bottom Surface Area = πr^2
- Total Surface Area
= L + T + B = $2\pi r h + 2(\pi r^2) = 2\pi r(h+r)$

Radius: 4
Height: 6

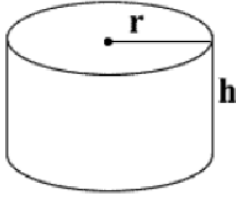
Calculate

Result of Circular Cylinder Shape

Volume = 301.59289474462014
Lateral Surface Area = 150.79644737231007
Top Surface Area = 50.26548245743669
Bottom Surface Area = 50.26548245743669
Total Surface Area = 251.32741228718345

OK

Circular Cylinder



- Volume = $\pi r^2 h$
- Lateral Surface Area = $2\pi r h$
- Top Surface Area = πr^2
- Bottom Surface Area = πr^2
- Total Surface Area
= L + T + B = $2\pi r h + 2(\pi r^2) = 2\pi r(h+r)$

Radius: 4
Height: 00

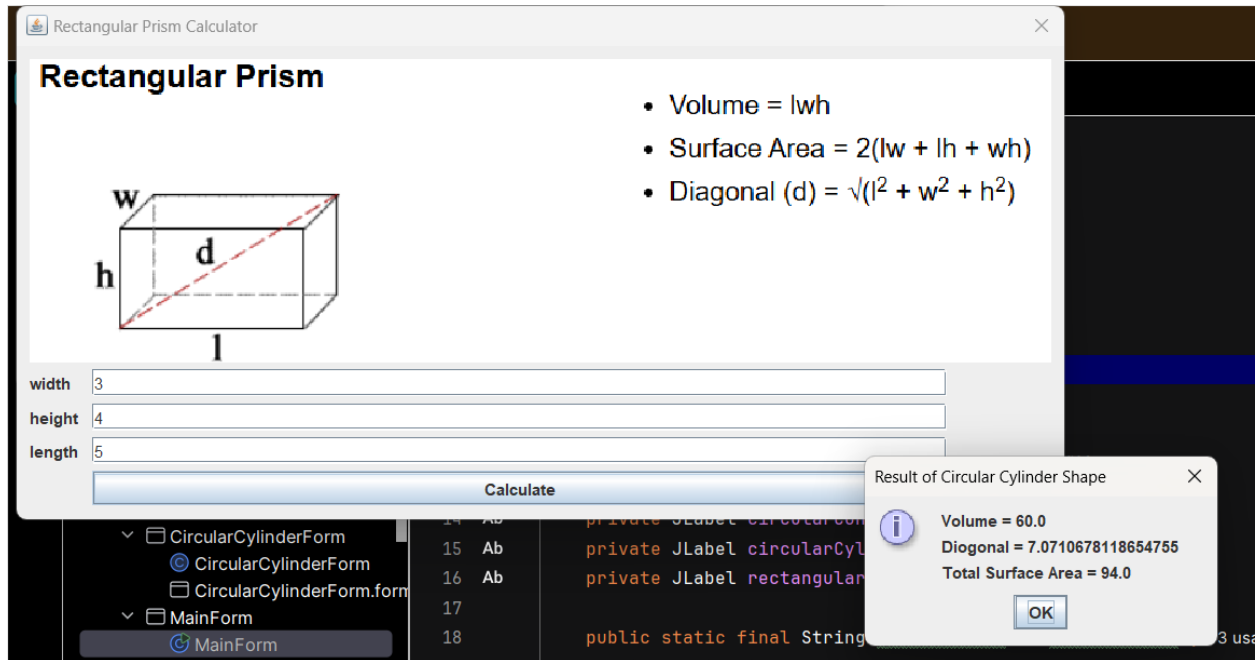
Calculate

Error

Please enter number Only!

OK

Rectangular Prism



The screenshot shows the 'Rectangular Prism Calculator' window. It features a diagram of a rectangular prism with dimensions labeled: width (w), height (h), length (l), and diagonal (d). The formulas for Volume, Surface Area, and Diagonal are listed. The input fields are filled with: width = 3, height = 4, and length = 5. The 'Calculate' button is highlighted. A small dialog box titled 'Result of Circular Cylinder Shape' is open, displaying the results: Volume = 60.0, Diagonal = 7.0710678118654755, and Total Surface Area = 94.0. The background shows a Java IDE with a project structure and code snippets.

Rectangular Prism

- Volume = lwh
- Surface Area = $2(lw + lh + wh)$
- Diagonal (d) = $\sqrt{l^2 + w^2 + h^2}$

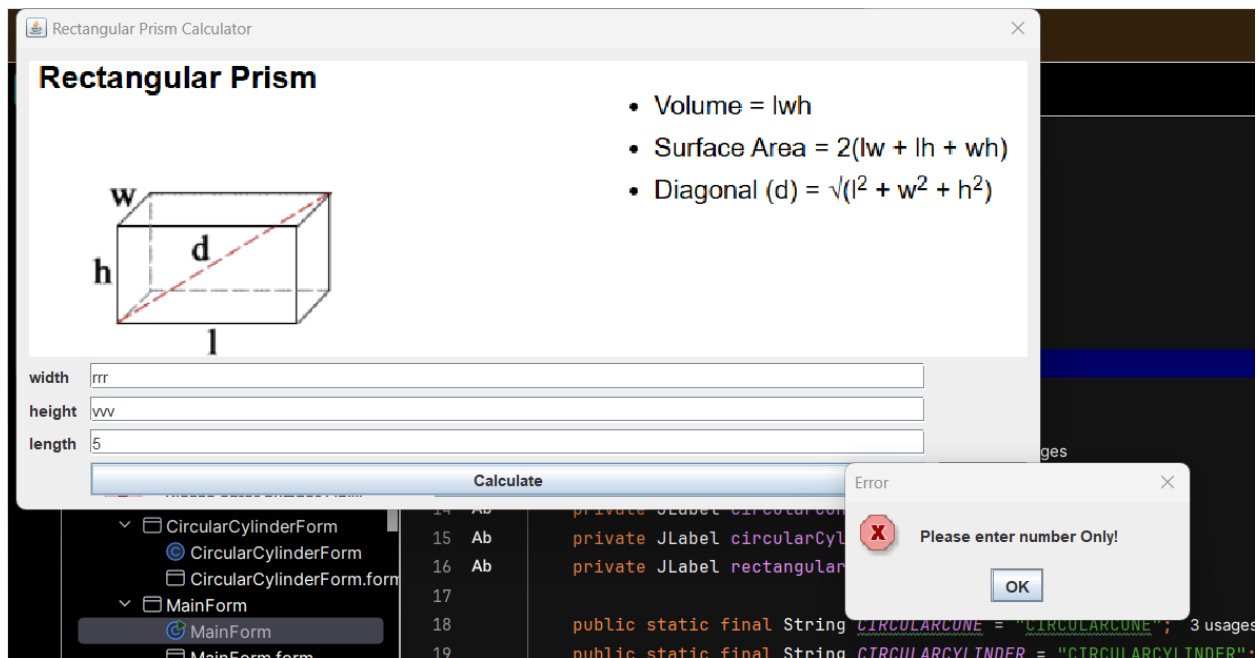
width: 3
height: 4
length: 5

Calculate

Result of Circular Cylinder Shape

Volume = 60.0
Diagonal = 7.0710678118654755
Total Surface Area = 94.0

OK



The screenshot shows the 'Rectangular Prism Calculator' window. The input fields are filled with: width = rrr, height = vv, and length = 5. The 'Calculate' button is highlighted. An error dialog box titled 'Error' is open, displaying the message: 'Please enter number Only!'. The background shows a Java IDE with a project structure and code snippets.

Rectangular Prism

- Volume = lwh
- Surface Area = $2(lw + lh + wh)$
- Diagonal (d) = $\sqrt{l^2 + w^2 + h^2}$

width: rrr
height: vv
length: 5

Calculate

Error

Please enter number Only!

OK